

PALLAVARAJAN T

AI & Machine Learning Engineer

Pallavarajan2005@gmail.com · 8754910894 · Pudukkottai, Tamil Nadu · [LinkedIn](#) · [GitHub](#)

EXECUTIVE SUMMARY

Computer Science and Engineering student with hands-on experience building real-world AI applications, including a YOLOv11-based object detection system, an LSTM text generator trained from scratch, and an NLP-powered resume matching tool using Sentence Transformers and FastAPI. Comfortable working across the full AI development stack, from training models in PyTorch to deploying production-ready APIs. Currently deepening my expertise in Transformers and applied NLP

TECHNICAL SKILLS

Languages	Python, SQL
AI / ML	TensorFlow, FastAPI, PyTorch, scikit-learn, OpenCV, NumPy, Pandas
NLP	Sentence-Transformers, Transformers, Hugging Face
Tools & Platforms	Git, GitHub, VS Code
Core Concepts	Deep Learning, Computer Vision, NLP

PROJECT

ResumeMatch AI

github.com/ResumeMatch AI

Python, FastAPI, Sentence-Transformers (all-MiniLM-L6-v2), HTML/JS

- Built a resume evaluation web app with a FastAPI backend, using sentence-transformer embeddings and cosine similarity for semantic resume-to-job-description matching, layered with regex-based skill extraction and experience parsing.
- Iterated the pipeline from an initial TF-IDF + scikit-learn matching approach to a semantic sentence-transformer model for better accuracy; built a single-file HTML/JS frontend for the matching interface.

Social Distancing Detector (YOLOv11)

github.com/pallava-06/Social-distance-monitoring-system

Python, YOLOv11 (Ultralytics), OpenCV, NumPy, SciPy

- Built a real-time computer vision pipeline that detects people in video footage using YOLOv11 and flags social distancing violations by computing pairwise Euclidean distances between detection centroids (via `scipy.spatial.distance.cdist`), with configurable confidence and distance threshold.
- Implemented live annotated video output — green/red bounding boxes for safe vs. violating pairs, a running violation counter, and exported the fully processed video — with a live preview mode for real-time monitoring.

INTERNSHIP

Applied Machine Learning in Python

10 DEC 2025 – 30 DEC 2025

VJ Infotech [Salem]

- Worked with Python libraries (NumPy, Pandas, and Scikit-learn) to implement core Machine Learning concepts through hands-on projects.

Deep Learning

Pantech Prolabs India Pvt Ltd [Remote]

- Completed Deep Learning internship covering neural network fundamentals
- Perceptrons, activation functions, gradient descent, and backpropagation with hands-on implementation.

EDUCATION

B.E CSE

2023 – 2027

Amrita college of engineering and technology — Nagercoil | CGPA: 8.14

CERTIFICATIONS

- Applied Machine Learning in python — VJ Infotech, 2025